1.Severity Distribution:

Low	1988
Medium	2464
High	1653

2.Category & Severity Distribution Table:

Category	**High**	**Low**	**Medium**	**Total**
Electricity	206	380	421	1007
Food	168	393	475	1036
Healthcare	342	294	379	1015
Road & Transport	387	230	444	1061
Waste	214	369	432	1015
Water	336	322	313	971
Total	**1653**	**1988**	**2464**	**6105**

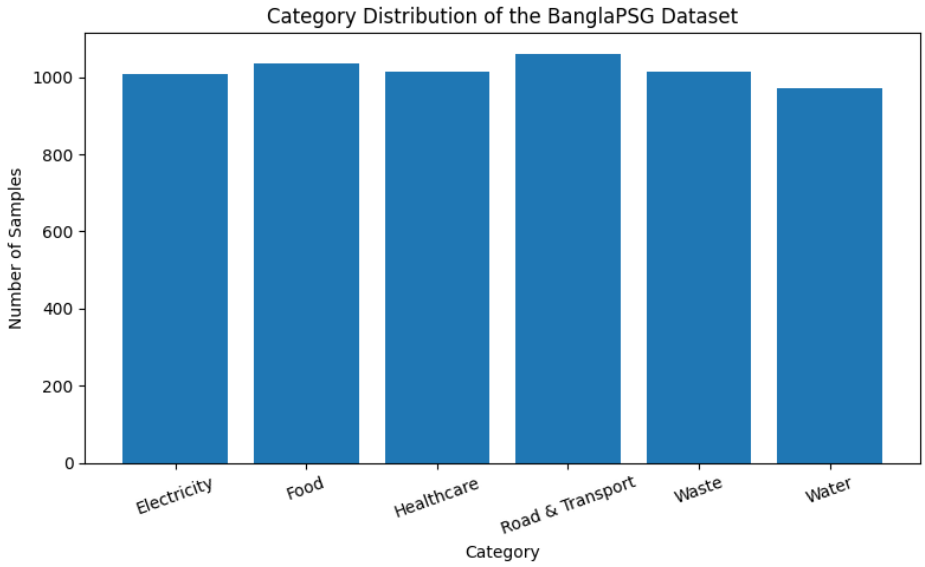
3.Dataset Statistics:

Total samples	6105
Vocabulary size	20804
Unique Words	20804
Average words	21.41
Median Words	18.0
Average Characters	125.78
Maximum Characters:	762
Minimum Characters:	8
Standard Deviation (words)	13.53

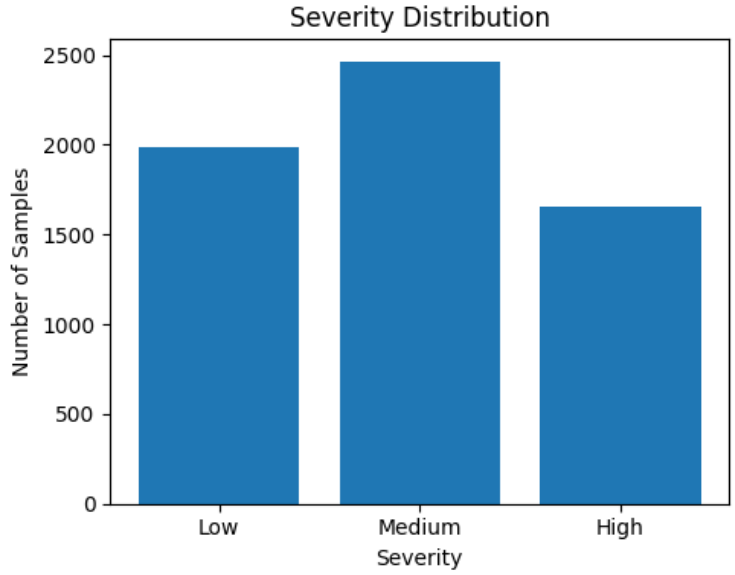
4.Class imbalance analysis:

The dataset is relatively well balanced. The number of samples per category ranges from 971 (Water) to 1,061 (Road & Transport) indicating no severe class imbalance. The slightly higher number of Road & Transport complaints may reflect the greater frequency of public discussions on transportation and road infrastructure in online platforms, while the relatively lower number of Water-related complaints may be associated with the availability of fewer publicly shared grievance posts during the data collection period. These variations are likely influenced by the natural distribution of publicly available social media content rather than intentional sampling decisions.

Bar chart for Category:



Bar chart for severity :



Baseline Models:

1. SVM model using –

CATEGORY
===== Validation Results =====
Accuracy: 0.7194323144104804
Precision: 0.7218738263492255
Recall: 0.7194323144104804
F1-score: 0.720338145077272
===== Test Results (Final) =====
Accuracy: 0.6932314410480349
Precision: 0.6953931076274006
Recall: 0.6932314410480349
F1-score: 0.6939655497728138
SEVERITY:
===== Validation Results =====
Accuracy: 0.4115720524017467
Precision: 0.4087181370430641
Recall: 0.4115720524017467
F1-score: 0.4094416390339834

===== Test Results (Final) =====
Accuracy: 0.3864628820960699
Precision: 0.38386849683414054
Recall: 0.3864628820960699
F1-score: 0.38361390593211686
]

	Algorithm	Category (validation)	Category (Test)	Severity (Validation)	Severity (Test)
SVM model	Accuracy	0.719 = 71.9	0.693 = 69.3%	0.412	0.386
	Precision	0.721 = 72.1%	0.695 = 69.5%	0.409	0.384
	Re-call	0.719 = 71.9%	0.693 = 69.3%	0.412	0.386
	F1 - score	0.720=72.0%	0.694 = 69.4%	0.409	0.384

Classification Report:

	Precision	Recall	F1-score	Support
Electricity	0.73	0.67	0.70	151
Food	0.66	0.65	0.65	156
Healthcare	0.78	0.78	0.78	152
Road & Transport	0.70	0.70	0.70	159
Waste	0.76	0.76	0.76	153
Water	0.55	0.59	0.57	145

	Precision	Recall	F1-score	Support
High	0.35	0.29	0.32	248
Medium	0.41	0.46	0.43	369
Low	0.38	0.38	0.38	299

category	Precision	Recall	F1-score	Support
Accuracy			0.69	916
macro avg	0.69	0.69	0.69	916
weighted avg	0.70	0.69	0.69	916

Severity	Precision	Recall	F1-score	Support
Accuracy			0.39	916
macro avg	0.38	0.38	0.38	916
weighted avg	0.38	0.39	0.38	916

Confusion Matrix:

	Elec	Food	Health	Road	Waste	Water
Electricity	[101	12	8	11	4	15]
Food	[9	101	9	10	4	23]
Healthcare	[4	8	119	6	5	10]
Road&Trans	[11	11	5	112	9	11]
Waste	[4	4	5	12	116	12]
Water	[10	18	7	9	15	86]

Confusion Matrix:

	High	Low	Medium
High	[71	70	107]
Low	[46	115	138]
Medium	[83	118	168]

2. BanglaBert:

Category:

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	1.260058	0.839247	0.924672	0.926650	0.923894	0.924105
2	0.464675	0.493222	0.936681	0.936398	0.936170	0.936055
3	0.312755	0.516359	0.936681	0.937409	0.935902	0.936130

{'eval_loss': 0.5884544849395752,
'eval_accuracy': 0.9301310043668122,
'eval_precision': 0.932144973477472,
'eval_recall': 0.9294871073253835,
'eval_f1': 0.9301213589713648,
'eval_runtime': 5.2425,
'eval_samples_per_second': 174.724,
'eval_steps_per_second': 11.063,
'epoch': 3.0}

	precision	recall	f1-score	support
Electricity	0.96	0.95	0.95	151
Food	0.94	0.95	0.94	156
Healthcare	0.96	0.95	0.95	152
Road & Transport	0.90	0.94	0.92	159
Waste	0.88	0.93	0.91	153

Water	0.96	0.86	0.91	145
accuracy			0.93	916
macro avg	0.93	0.93	0.93	916
weighted avg	0.93	0.93	0.93	916

Severity:

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.113176	2.165750	0.408297	0.276494	0.346514	0.259094
2	2.004215	2.152412	0.425764	0.418829	0.380240	0.347468
3	1.828205	2.183532	0.442140	0.440293	0.403449	0.386391


```
{'eval_loss': 2.141801357269287,
 'eval_accuracy': 0.43013100436681223,
 'eval_precision': 0.4329999140882232,
 'eval_recall': 0.39418328260133756,
 'eval_f1': 0.37889604383528236,
 'eval_runtime': 5.605,
 'eval_samples_per_second': 163.426,
 'eval_steps_per_second': 10.348,
 'epoch': 3.0}
```


	precision	recall	f1-score	support
High	0.41	0.14	0.21	248
Low	0.48	0.36	0.41	299
Medium	0.42	0.68	0.52	369
accuracy			0.43	916
macro avg	0.43	0.39	0.38	916
weighted avg	0.43	0.43	0.40	916

3. XLM-R:

Category:

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.954735	0.679244	0.917031	0.921224	0.915927	0.915989
2	0.518729	0.523550	0.936681	0.937474	0.935896	0.936146
3	0.278445	0.566249	0.934498	0.935203	0.933734	0.933815

```
{'eval_loss': 0.5551726222038269,
 'eval_accuracy': 0.9268558951965066,
 'eval_precision': 0.9280510721072162,
 'eval_recall': 0.9264828018168833,
 'eval_f1': 0.9267306661858554,
 'eval_runtime': 9.2801,
 'eval_samples_per_second': 98.706,
 'eval_steps_per_second': 6.25,
 'epoch': 3.0}
```

	precision	recall	f1-score	support
Electricity	0.97	0.96	0.97	151
Food	0.93	0.95	0.94	156
Healthcare	0.95	0.95	0.95	152
Road & Transport	0.89	0.89	0.89	159
Waste	0.88	0.95	0.91	153
Water	0.94	0.86	0.90	145
accuracy			0.93	916
macro avg	0.93	0.93	0.93	916
weighted avg	0.93	0.93	0.93	916

Severity:

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.180576	2.165743	0.403930	0.134643	0.333333	0.191809
2	2.155023	2.146981	0.402838	0.272980	0.335697	0.218740
3	2.119173	2.129627	0.422489	0.298132	0.364974	0.294446

```
{'eval_loss': 2.1123785972595215,
 'eval_accuracy': 0.4279475982532751,
 'eval_precision': 0.3032508843427619,
 'eval_recall': 0.3714519642409356,
 'eval_f1': 0.3023399014778325,
 'eval_runtime': 8.6745,
 'eval_samples_per_second': 105.597,
 'eval_steps_per_second': 6.686,
 'epoch': 3.0}
```

	precision	recall	f1-score	support
High	0.00	0.00	0.00	248
Low	0.50	0.27	0.35	299
Medium	0.41	0.84	0.55	369
accuracy			0.43	916
macro avg	0.30	0.37	0.30	916

```
weighted avg       0.33      0.43      0.34       916
```

4. mBERT:

Category:

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	1.290015	1.047053	0.837336	0.853530	0.838328	0.841505
2	0.748023	0.661231	0.906114	0.906049	0.905938	0.905919
3	0.458593	0.672563	0.913755	0.915432	0.913416	0.913638

```
{'eval_loss': 0.6910259127616882,
 'eval_accuracy': 0.9093886462882096,
 'eval_precision': 0.9124707219006698,
 'eval_recall': 0.9088989033981316,
 'eval_f1': 0.9098952162915156,
 'eval_runtime': 6.5809,
 'eval_samples_per_second': 139.19,
 'eval_steps_per_second': 8.813,
 'epoch': 3.0}
```

```
                   precision    recall  f1-score   support

     Electricity       0.96      0.93      0.94       151
            Food       0.91      0.95      0.93       156
      Healthcare       0.97      0.90      0.94       152
Road & Transport       0.84      0.92      0.88       159
           Waste       0.89      0.88      0.88       153
           Water       0.91      0.88      0.89       145

        accuracy                           0.91       916
       macro avg       0.91      0.91      0.91       916
    weighted avg       0.91      0.91      0.91       916
```

Severity:

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.140514	2.176309	0.410480	0.277769	0.350493	0.269923
2	2.081059	2.125903	0.438865	0.468080	0.392048	0.360526
3	1.914144	2.213595	0.397380	0.402257	0.389146	0.390111

```
{'eval_loss': 2.154573440551758,
 'eval_accuracy': 0.4148471615720524,
 'eval_precision': 0.4175916263180082,
 'eval_recall': 0.407311106277172,
```

```
 'eval_f1': 0.4088221494513227,
 'eval_runtime': 6.9192,
 'eval_samples_per_second': 132.385,
 'eval_steps_per_second': 8.382,
 'epoch': 3.0}
```

```
                precision    recall  f1-score   support

          High       0.35      0.38      0.36       248
           Low       0.48      0.36      0.42       299
        Medium       0.42      0.48      0.45       369

      accuracy                           0.41       916
     macro avg       0.42      0.41      0.41       916
  weighted avg       0.42      0.41      0.41       916
```

Overall

Category:

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
SVM	69.32%	69.54%	69.32%	69.39%
mBERT	90.94%	91.25%	90.89%	90.99%
XLM -R	92.68%	92.80%	92.65%	92.67%
BanglaBert	93.01%	93.21%	92.94%	93.01%

Severity:

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
SVM	38.65%	38.39%	38.65%	38.36%
mBERT	41.48%	41.76%	40.73%	40.88%
XLM -R	42.79%	30.32%	37.14%	30.23%
BanglaBert	43.01%	43.29%	39.42%	37.89%